Software Engineering Assignment (CO311)

B.Tech CSE 5th Sem Autumn 2025

Topic: Problem statement on Hospital Management System

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HOSPITAL MANAGEMENT SYSTEM

PATIENT MANAGEMENT IN HOSPITAL

Patient management in a hospital is a complex, systematic process designed to ensure that every patient receives safe, effective, and efficient healthcare from the moment they enter the hospital until after they are discharged. It is a multidisciplinary effort involving clinical, administrative, and support staff.

Problems in Traditional Patient Management:

Traditional processes consist of a series of disconnected, manual steps. A paper file is created at registration; doctors write notes by hand; nurses maintain a separate chart; laboratory requests are sent through physical forms; and billing staff manually compile charges later. This fragmented approach keeps information siloed, slows communication, and increases the risk of errors.

Requirements in Modern Patient Management:

1.Patient Portal: A secure portal/app for patients to book, reschedule, or cancel appointments, pre-register by filling out forms online to reduce waiting time, View lab results, discharge summaries, and prescriptions, make online payments and communicate securely with the care team.

2.Digital Front Desk: Queue Management System to display wait times and manage patient flow, Automated insurance verification, Digital consent forms.

3.Centralized Patient Record**:** A comprehensive digital chart accessible to authorized staff, containing demographics, medical history, allergies, medications, test results, and treatment plans.

4. Computerized Physician Order Entry (CPOE): Allows doctors to enter orders for medications, lab tests, and imaging electronically, reducing handwriting errors.

5. Clinical Documentation**:** Templates for admission notes, progress notes, and consultation reports that are easy for doctors and nurses to use.

6. Digital Task Management**:** A centralized list of tasks for each patient (e.g., "administer IV antibiotic at 4 PM," "check blood sugar")

7. Bed Management Dashboard**:** A real-time, visual overview of all beds in the hospital—showing which are occupied, available, or being cleaned.

8**.**Electronic Medication Administration Record (eMAR): Nurses can scan a patient's wristband and the medication's barcode to ensure the "Five Rights" of medication administration (Right Patient, Right Drug, Right Dose, Right Route, Right Time).

9.Laboratory Information System (LIS) Integration**:** Seamlessly place lab orders and receive results directly into the Central Patient Record being maintained.

10.Radiology Information System (RIS) & PACS Integration**:** Order imaging studies and allow clinicians to view images (X-rays, CTs) from within the patient's chart.

11.Billing Integration**:** Automatically capture charges for procedures, tests, and medications as they are ordered and administered, ensuring accurate and timely billing.

Software Requirements Specification (SRS)

Project: Digital Front Desk (DFD) — Private Hospitals (India)

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Date: 05/10/2025

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1. Introduction

1.1 Purpose

This SRS describes the functional and non-functional requirements for the **Digital Front Desk (DFD)** system intended for private hospitals in India. The document follows IEEE-style SRS conventions and is intended to be used by product managers, developers, QA, integration engineers, and deployment teams.

1.2 Scope

The DFD is a receptionist-focused module that provides appointment & OPD management, patient registration (OPD/IPD entry), token/queue management, billing initiation, insurance verification, basic EMR access, lab & pharmacy booking, and automated patient communications. The system integrates with an existing Hospital Information System (HIS), third-party LIS/PIS, payment gateways, and NDHM/ABDM services.

1.3 Audience

- Product Owner
- Project Manager
- Developers & Integration Engineers
- QA/Testers
- Deployment/Operations
- Hospital IT & Reception staff

1.4 Definitions, Acronyms, Abbreviations

- DFD: Digital Front Desk
- HIS: Hospital Information System
- MRN: Medical Record Number
- ABHA / Health ID: NDHM patient identifier
- HL7 / FHIR: Interoperability standards
- NABH: National Accreditation Board for Hospitals
- NDHM / ABDM: National Digital Health Mission / Ayushman Bharat Digital Mission
- API: Application Programming Interface

1.5 References

- Company integration guidelines
- Hospital's existing HIS API docs
- NDHM / ABDM interoperability guidelines

- NABH IT requirements

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2. Overall Description

2.1 Product Perspective

The DFD is a modular, web-based application (responsive) used by receptionists on desktops or tablets. It is not a full HIS; instead it integrates and synchronizes with the hospital's HIS to perform front-desk tasks and to avoid duplicate data entry.

2.2 Product Functions (High level)

- Appointment & OPD management
- Walk-in registration and token issuance
- OPD/IPD registration and MRN lifecycle
- Billing initiation (preliminary) and payment capture
- Insurance & scheme verification and claim initiation
- Basic EMR access (view-only at front desk) and link to HIS
- Lab & pharmacy order placement (triggering LIS/PIS workflows)
- Queue display & token announcement
- SMS/WhatsApp/email notifications
- Reports and dashboards for reception managers

2.3 User Classes and Characteristics

Role	Primary Tasks	Technical Proficiency
Reception ist	Register patients, issue tokens, initiate billing, manage queues	Basic computer literacy
Front-desk Superviso r	Configure schedules, view reports, manage staff	Intermediate
HIS Admin / Integratio n Engineer	Configure connectors, monitor synchronization	Advanced
Admin / Finance	Reconcile payments, export MIS	Advanced

2.4 Operating Environment

- Server: Linux (Ubuntu 24.04 LTS recommended) or cloud provider region within India
- Backend: Node.js / Java / .NET (TBD by project)
- Database: PostgreSQL / MySQL
- Client: Chrome/Edge browsers on desktop/tablet, touch screen support
- Connectivity: HTTPS-based connections to HIS, NDHM, payment gateways

2.5 Design & Implementation Constraints

- Must integrate with the hospital's existing HIS; cannot assume HIS replacement
- Must use supported standards (HL7/FHIR) where HIS supports them
- Must maintain logs/audit trails for regulatory audits
- Data retention & residency within India

2.6 Assumptions & Dependencies

- HIS provides an API or middleware for integration
- OCR/scanning devices for ABHA QR or ID cards are available
- Payment gateway and SMS/WhatsApp providers are provisioned by hospital

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3. Specific Requirements

3.1 Functional Requirements (by module)

Each functional requirement is uniquely identified (FR-###), includes priority, description, and acceptance criteria.

3.1.1 Appointment & OPD Management

- **FR-001 — Appointment Creation (High):** System shall allow receptionist to create, reschedule, cancel appointments and mark walk-ins. *Acceptance:* Appointment reflected in DFD and synced to HIS within 5 seconds.
- **FR-002 — Appointment View (High):** System shall display day's schedule with filters by doctor, specialty, and time slots.
- **FR-003 — No-show Handling (Medium):** Mark no-shows and optionally trigger follow-up reminders.

3.1.2 Patient Registration (OPD/IPD)

- **FR-010 — New Patient Registration (High):** Capture demographics, contact, ID proofs, assign MRN, and optionally link/generate ABHA. *Acceptance:* MRN created and visible in HIS; ABHA validation returns success or clear error.

- **FR-011 — Update Patient Details (Medium):** Edit non-critical fields with audit trail.
- **FR-012 — IPD Admission Initiation (Medium):** For admissions, capture bed/ward, admitting consultant, provisional diagnosis; push admission ADT message to HIS.

3.1.3 Queue & Token Management

- **FR-020 — Token Generation (High):** Auto-generate sequential tokens for appointments and walk-ins; support QR and printed token. *Acceptance:* Token number issued and queue updated in <2s.
- **FR-021 — Token Display & Announce (High):** Display current tokens on waiting-room screens; allow manual override and emergency priority.
- **FR-022 — SMS/WhatsApp Token Notification (Medium):** Send token number and estimated wait time when issued.

3.1.4 Billing Initiation

- **FR-030 — Provisional Bill Creation (High):** Create a provisional bill capturing consultation charges and optional lab/pharmacy pre-charges.
- **FR-031 — Multi-Mode Payment Capture (High):** Accept payments via cash, card, UPI, and record partial payments/deposits.
- **FR-032 — Integration with Billing Engine (High):** Push provisional bill to HIS/Billing module and reconcile real-time payment status.

3.1.5 Insurance & Scheme Verification

- **FR-040 — Insurance Eligibility Check (High):** Verify insurance / TPA eligibility via API. *Acceptance:* Eligibility response in <5s.
- **FR-041 — AB-PMJAY / Government Schemes (High):** Initiate required pre-authorization/claim initiation steps per scheme.

3.1.6 Basic EMR Access

- **FR-050 — EMR Lookup (Medium):** Search and view recent visits, outstanding reports, and allergies (read-only at front desk).

3.1.7 Lab & Pharmacy Integration

- **FR-060 — Lab/Pharmacy Order Placement (Medium):** Create lab test booking or pharmacy order and push to LIS/PIS.
- **FR-061 — Receive Test Status (Low):** Show test status (sample collected, result ready) from LIS.

3.1.8 Reporting & Dashboards

- **FR-070 — Daily Summary Report (High):** Generate counts for registrations, bill totals, no-shows, average wait time.
- **FR-071 — Export MIS (Medium):** Export reports to CSV/PDF.

3.1.9 Patient Communications

- **FR-080 — Notifications (High):** Send appointment reminders and token calls via SMS/WhatsApp/email.
- **FR-081 — Templates (Medium):** Admin can configure message templates and opt-in preferences.

3.2 Non-Functional Requirements

3.2.1 Usability

- **NFR-001:** Receptionist dashboard shall load within 1.5s on local LAN.
- **NFR-002:** UI shall support English and configurable regional languages.

3.2.2 Reliability & Availability

- **NFR-010:** System uptime >= 99.5% (excluding scheduled maintenance).

3.2.3 Security

- **NFR-020:** All traffic encrypted using TLS 1.2 or higher.
- **NFR-021:** Passwords hashed with strong algorithm (bcrypt/argon2).
- **NFR-022:** Role-based access control; audit logs for create/update/delete operations.

3.2.4 Performance

- **NFR-030:** Token issuance and queue update should reflect across all clients in under 2 seconds on LAN.
- **NFR-031:** System shall handle 100 concurrent receptionist sessions per hospital instance.

3.2.5 Maintainability & Extensibility

- **NFR-040:** Provide well-documented REST APIs and connector interfaces for HIS, LIS, and third-party services.

3.2.6 Localization & Formats

- **NFR-050:** Date/time formats default to DD/MM/YYYY and currency in INR. Support input of Aadhar/PAN IDs where needed.

3.3 External Interface Requirements

3.3.1 User Interfaces

- Responsive web UI (desktop/tablet)
- Waiting-room display screen (read-only) for token call
- Optional kiosk UI for self-check-in

3.3.2 Hardware Interfaces

- ABHA QR scanner
- Barcode/ID scanners
- Card reader (optional) for ID verification
- Printer for tokens/receipts

3.3.3 Software Interfaces

- HIS: HL7/FHIR/REST APIs for patient, appointment, billing
- NDHM / ABHA: OAuth2-secured REST endpoints for ABHA validation and PHR updates
- LIS/PIS: HL7/FHIR integration for orders/results
- Payment Gateway: Standard gateway APIs (PCI-DSS compliance recommended)
- SMS / WhatsApp gateway: REST APIs

3.4 Security & Privacy Requirements

- Patient consent must be recorded for any PII or clinical data exchange (per NDHM consent framework).
- Access control: least privilege for receptionist; elevated roles for finance/HIS admin.
- Audit trail for all data access/modification with timestamp, user-id, and IP.
- Data at rest encryption for sensitive fields (PHI) as per hospital policy.

3.5 Performance Requirements

- End-to-end ABHA validation should complete within 5 seconds under standard load.
- Background sync jobs (e.g., reconcile payments) may run asynchronously and must not block receptionist operations.

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4. System Architecture & Integration

4.1 Integration Standards

- HL7 v2.x for legacy HIS adapters (ADT, ORM, ORU)
- FHIR R4 for NDHM and modern HIS interactions (Patient, Encounter, Observation, MedicationRequest)
- OAuth2 and TLS for authentication/authorization to NDHM and third-party services
- JSON over HTTPS for REST APIs

4.2 Data Flows

- Registration -> MRN generation -> ABHA validation -> Push to HIS (ADT)
- Appointment booking -> Token generation -> Queue display -> Doctor call -> Billing initiation -> Finalize bill in HIS
- Lab order -> LIS -> Sample status -> Result posted -> EMR updated

4.3 API Contracts (summary)

Provide OpenAPI/Swagger specs for these endpoints (examples): - POST /api/patients (create/update patient) - GET /api/patients/{mrn} - POST /api/appointments - POST /api/tokens - POST /api/billing/provisional - POST /api/integrations/ndhm/validate-abha

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5. Workflows & Use Cases

5.1 Primary Workflows

1. **Walk-in Patient Registration & Token:** Receptionist scans ID/ABHA or enters data -> system creates MRN -> token issued -> queue updated -> SMS confirmation sent.
2. **Appointment Check-in:** Patient arrives, receptionist confirms appointment -> check-in -> token assigned -> patient waits.
3. **Billing Initiation:** After consultation, receptionist creates provisional bill -> captures payment partial/full -> pushes final bill to HIS accounting.

5.2 Use Case Matrix (high-level)

Use Case	Actor	Trigger	Precondition	Postcondition
Register patient	Receptionist	Walk-in / New appointment	HIS reachable	MRN created and stored
Issue token	Receptionist / Kiosk	Registration or check-in	MRN exists	Token issued; queue updated
Verify insurance	Receptionist	During billing	Provider TPA API available	Eligibility result obtained

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6. Data Model & Reports

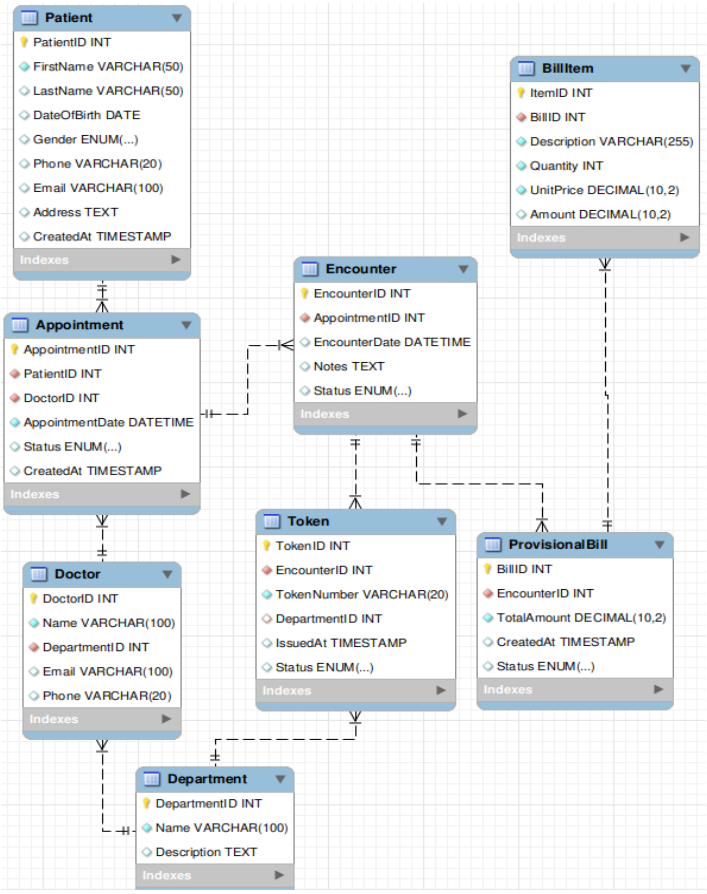
6.1 Key Data Entities

- Patient (MRN, name, contact, ABHA, demographics)

- Appointment (id, patient, doctor, timeslot, status)
- Token (token_no, type, status, issued_at)
- Encounter (visit id, department, consultant)
- ProvisionalBill (id, items, total, payments)
- LabOrder / PharmacyOrder

6.2 Sample ERD (textual & diagram)

- Patient 1 — * Appointment
- Appointment 1 — 1 Encounter
- Encounter 1 — * Token
- Encounter 1 — 1 ProvisionalBill



6.3 Standard Reports & Dashboards

- Daily front-desk summary (registrations, billed amount, no-shows)
- Average wait time per department
- Token throughput per hour
- Outstanding payments & deposits

7. Validation, Acceptance & Test Criteria

7.1 Acceptance Criteria (examples)

- **AC-01:** New patient registration successfully creates MRN in DFD and syncs to HIS within 5s.
- **AC-02:** Token issuance displays on all waiting-room screens within 2s.
- **AC-03:** ABHA validation returns status (valid/invalid) with error details when invalid.
- **AC-04:** Provisional bill can be created and a partial payment recorded; status reconciles with HIS billing.

7.2 Example Test Cases

- **TC-001:** Register patient with ABHA QR — expect MRN + ABHA link.
- **TC-002:** Create appointment for Dr. X at 10:30 — expect appointment visible in day view.
- **TC-003:** Issue token and verify waiting-room display updates.
- **TC-004:** Simulate insurance API failure — ensure graceful fallback and error message.

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8. Compliance, Legal & Localization

8.1 NDHM / ABDM Requirements

- ABHA generation / linking flow integrated at registration
- Consent capture and storage per NDHM consent artifact model
- Use NDHM FHIR profiles for PHR updates where applicable

8.2 NABH & Clinical Establishments Act

- Maintain audit trails, secure storage, and logs for required durations
- Support standard operating workflows required by NABH accreditation (consent, patient identification, incident logging)

8.3 Data Residency & Retention

- Production data stored in India (cloud region or on-premise)
- Retention policy: patient records retained per hospital policy/state law (e.g., 7 years typical; configurable)

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9. Deployment & Operational Requirements

9.1 Deployment Topologies

- **Single-site deployment:** On-premise servers with local LAN for front-desk clients; cloud connectors for NDHM
- **Cloud-hosted deployment:** All services hosted in India cloud region (private VPC) with secure VPN to HIS if HIS is on-prem.

9.2 Backup/Recovery & Monitoring

- Daily database backups + point-in-time recovery for last 7 days
- Monitoring (uptime, queue lengths, failed integrations) with alerting to hospital IT

9.3 Support & Maintenance

- 24x6 support SLA (configurable)
- Regular updates for NDHM/HL7/FHIR changes

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10. Appendices

10.1 Revision History

Version	Date	Author	Change
1.0	05/10/2025	Kishan Harlalka Rahul Yadav Ruknabh Bhattacharyya Raktim Ranjan Handique	Initial draft

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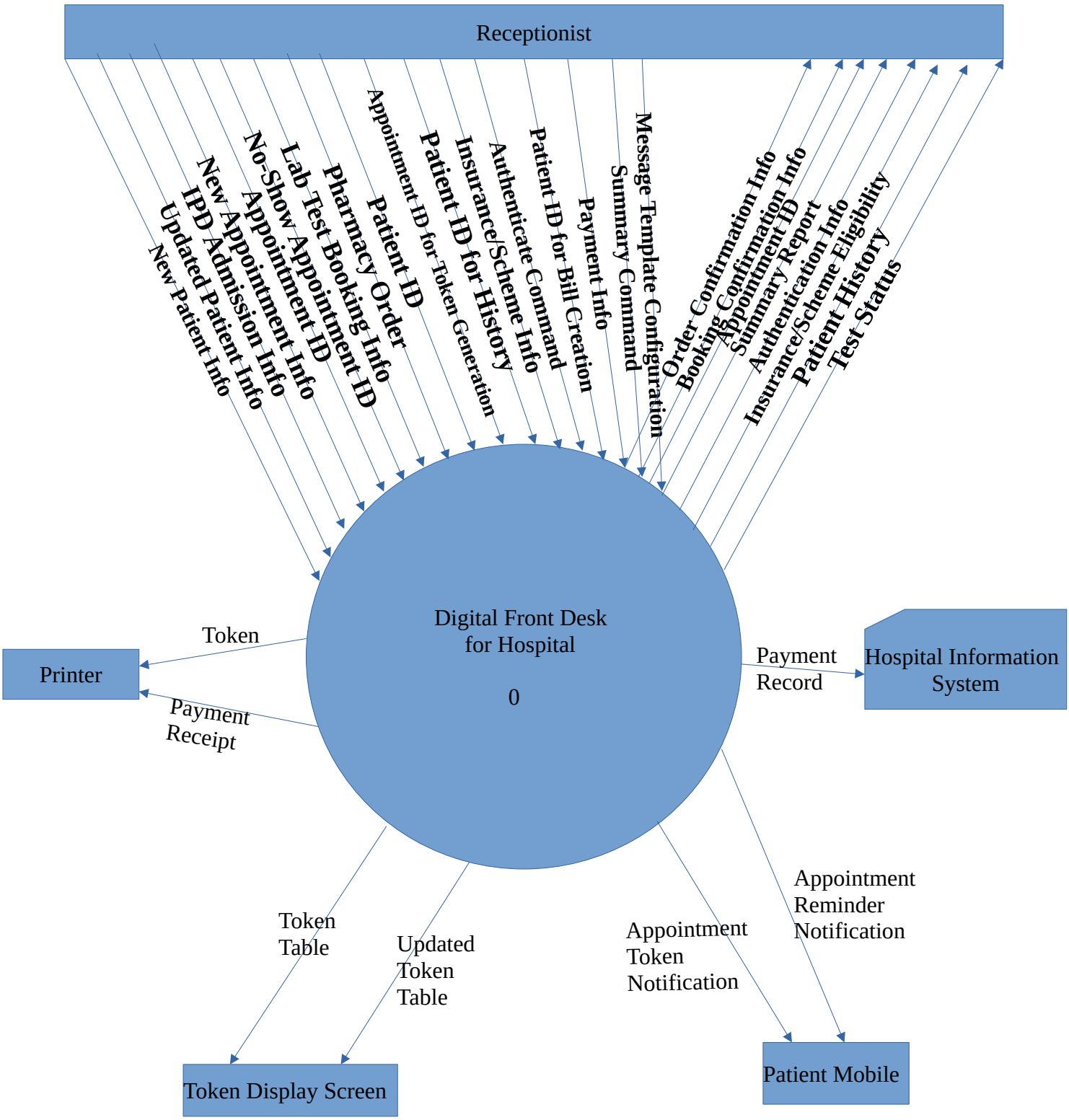
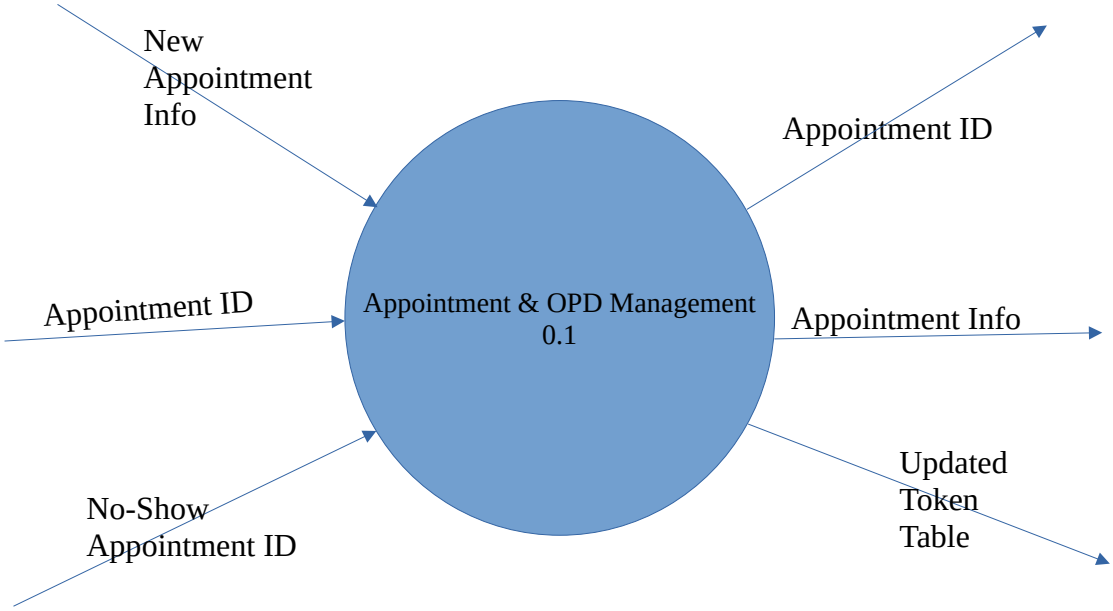
End of SRS Document

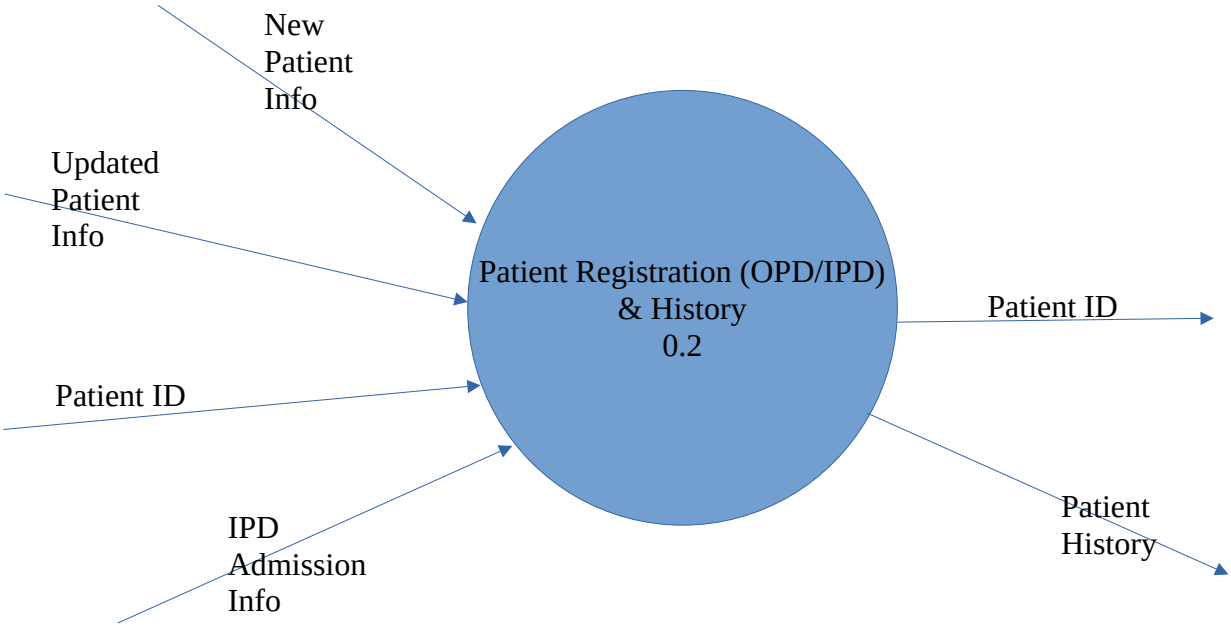
Design Document
(Hospital Digital Front Desk Management)

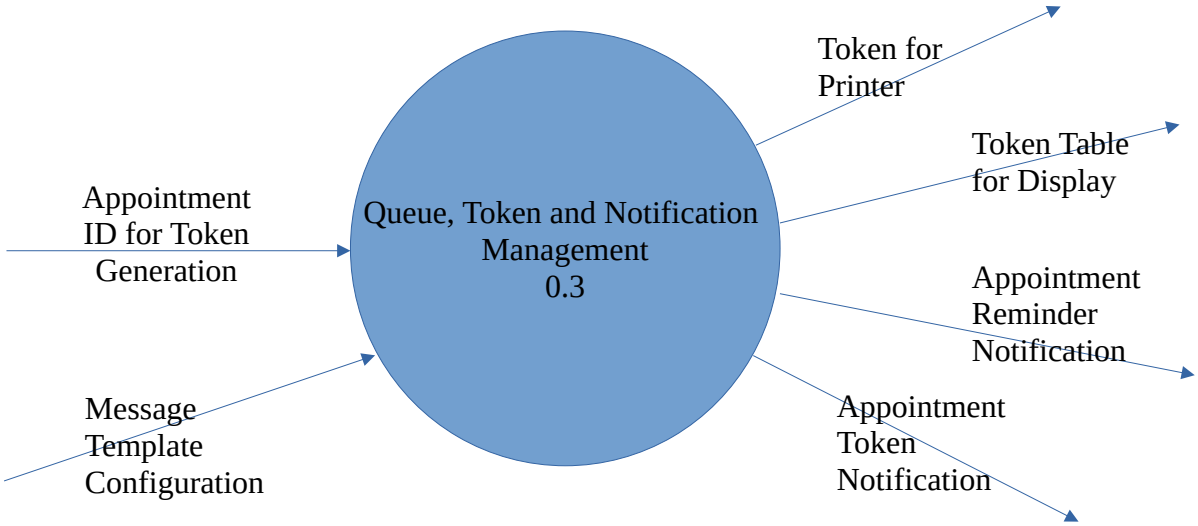
Level 0 Data Flow Diagram :

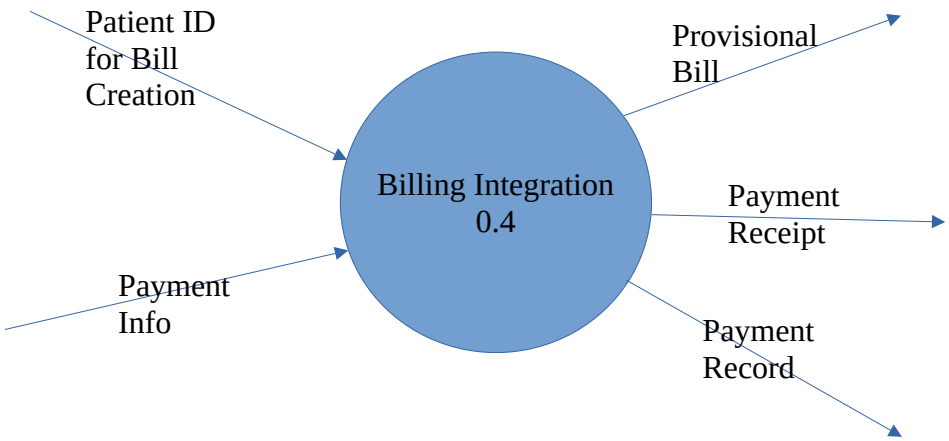


Level 1 Data Flow Diagrams :





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Provisional
Bill
Billing Integration
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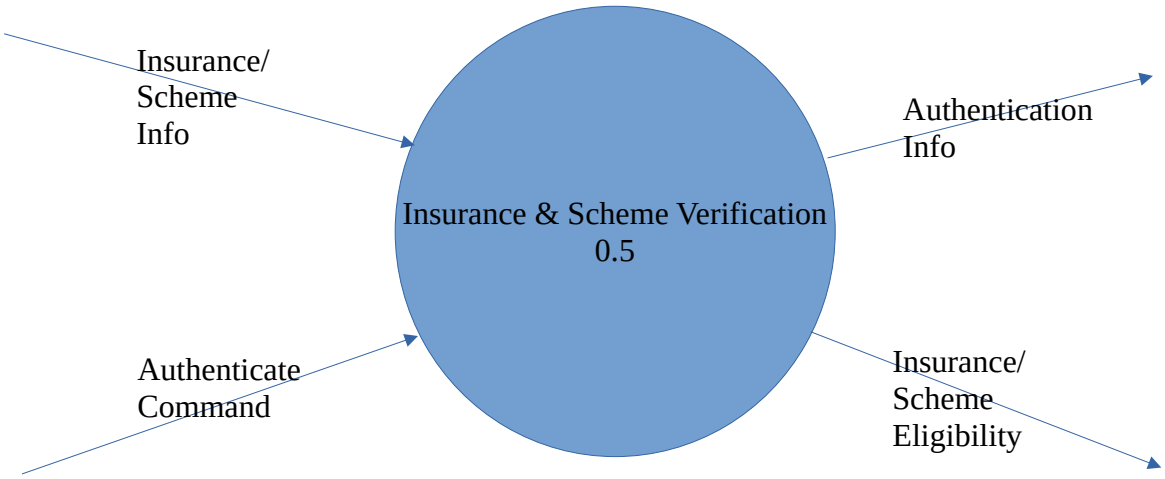
Payment
Info
Payment
Record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Authenticate
Command
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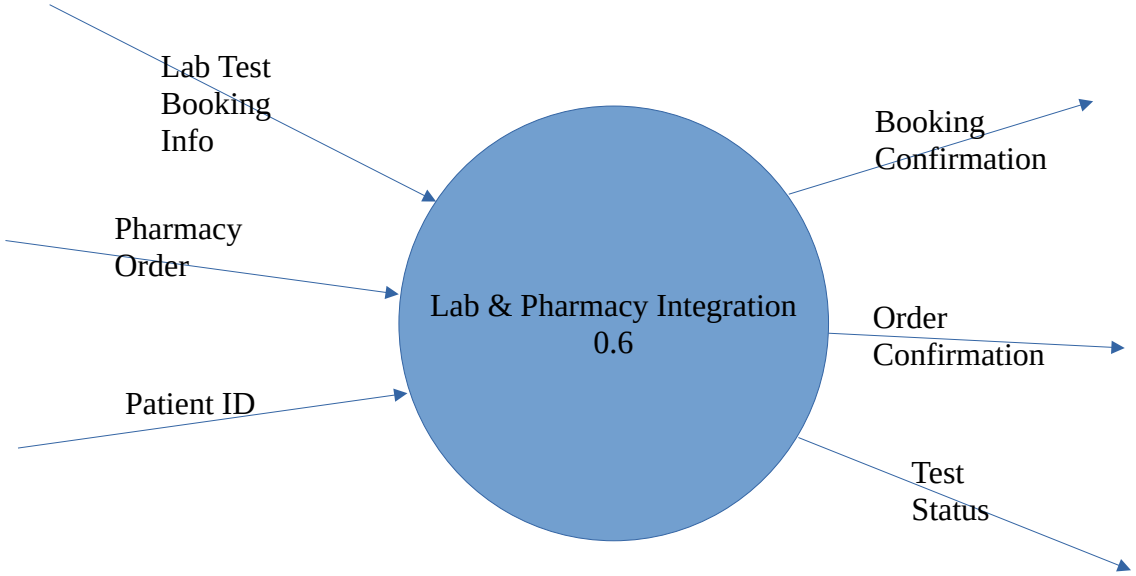
Scheme
Eligibility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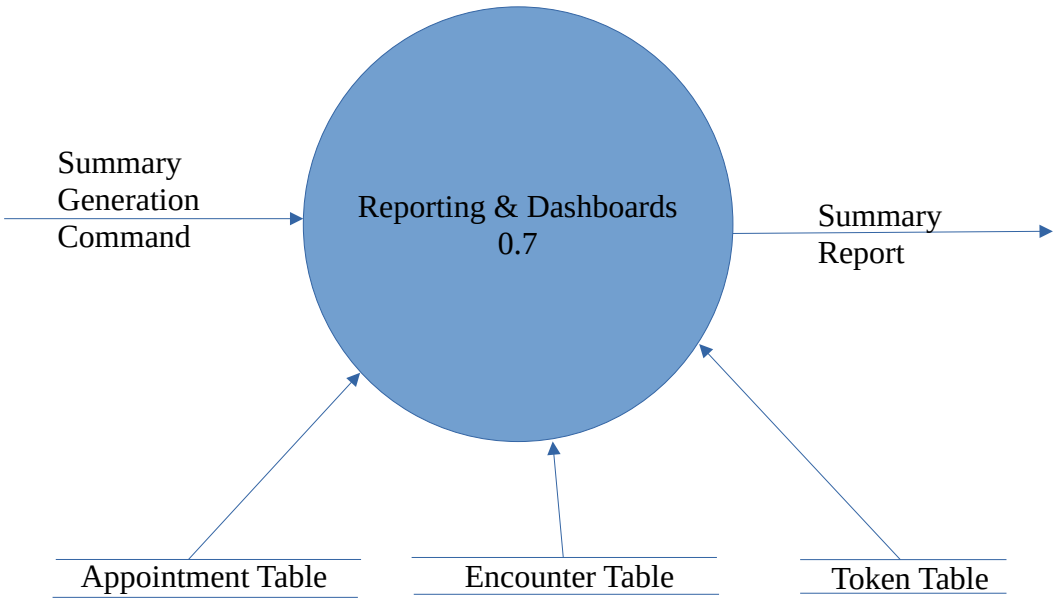
Order Confirmation
Test Status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Summary Report
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Encounter Table
Token Table

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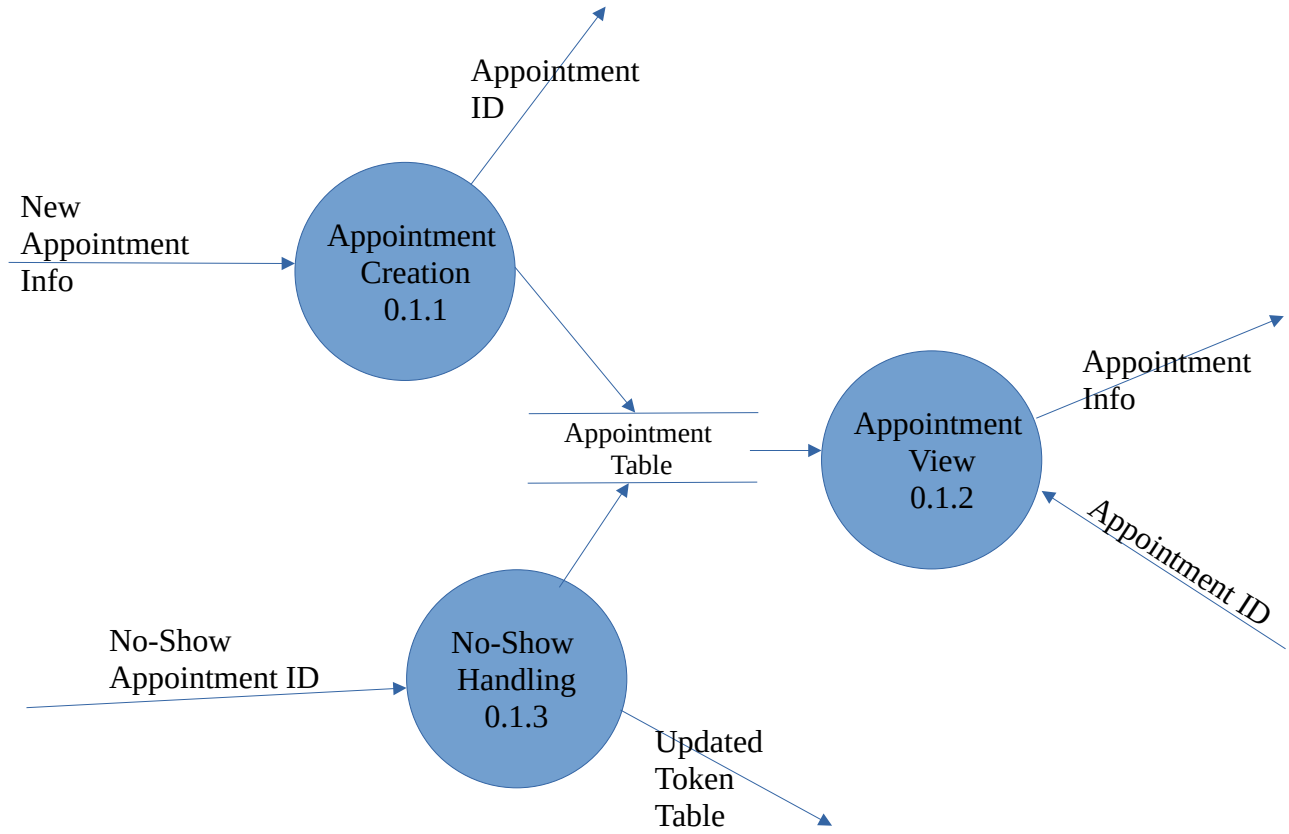

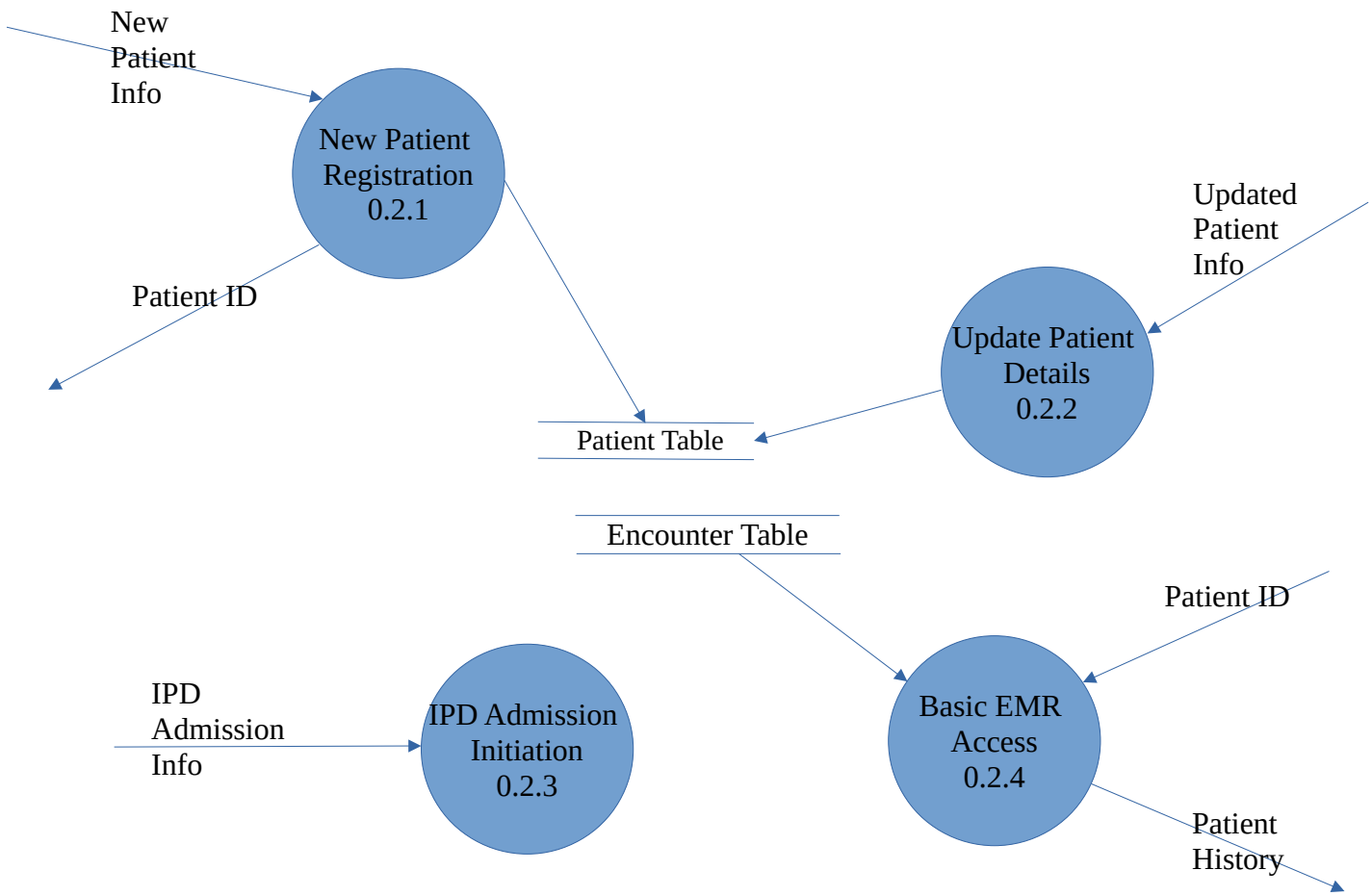

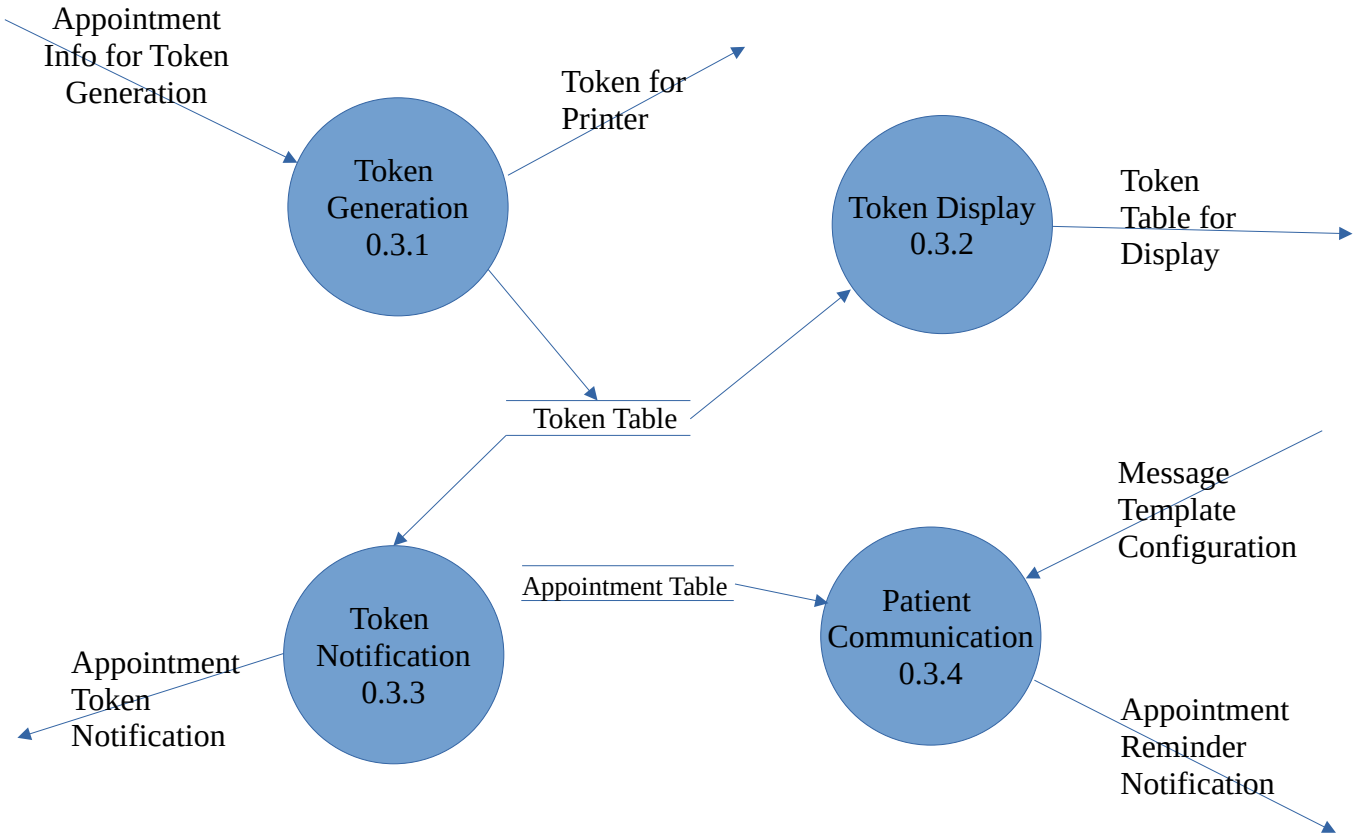
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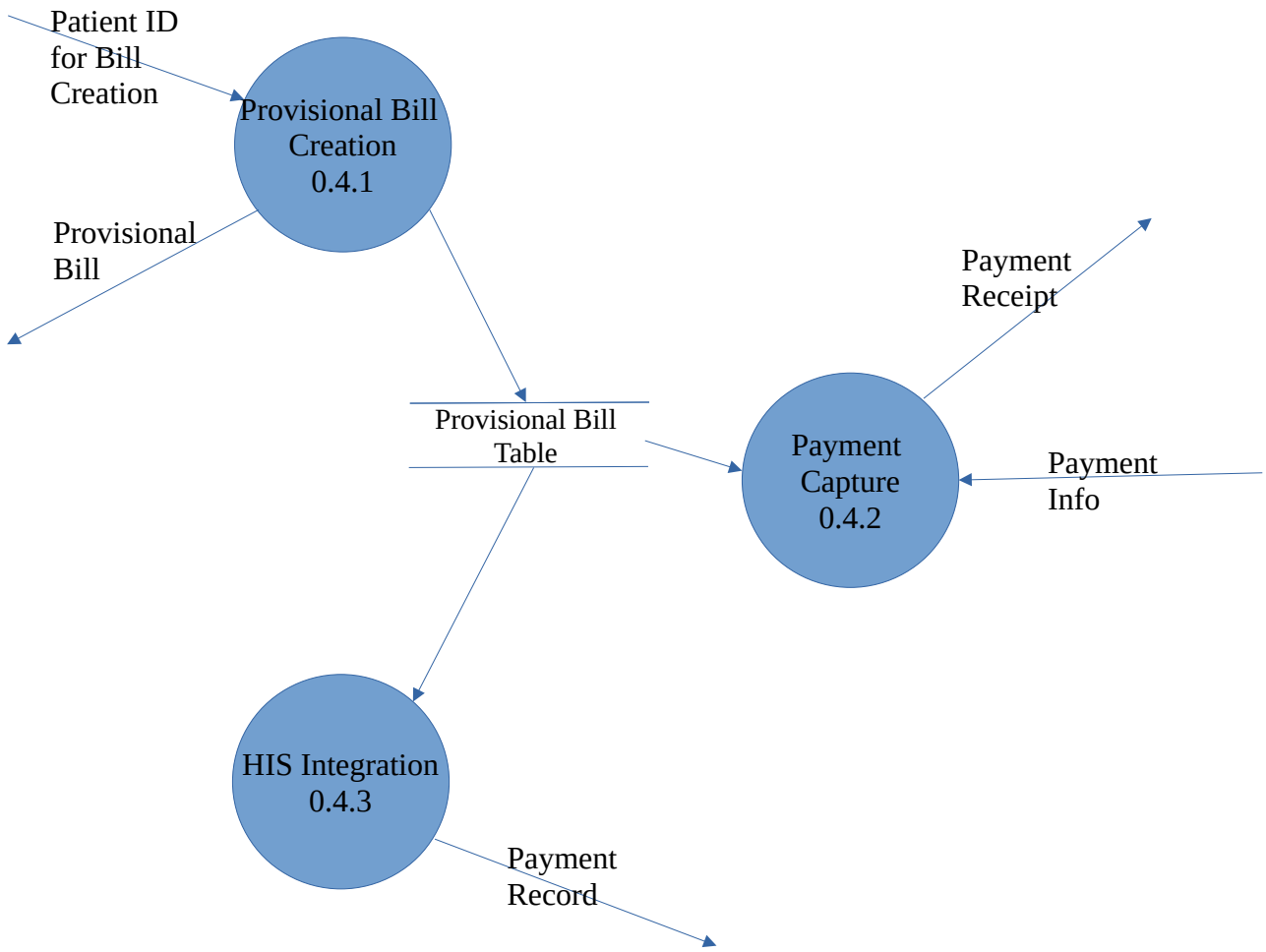
Token for Printer
Token Display 0.3.2
Token Table for Display
Token Table
Message Template Configuration
Appointment Table
Patient Communication 0.3.4
Token Notification 0.3.3
Appointment Token Notification
Appointment Reminder Notification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Payment Receipt
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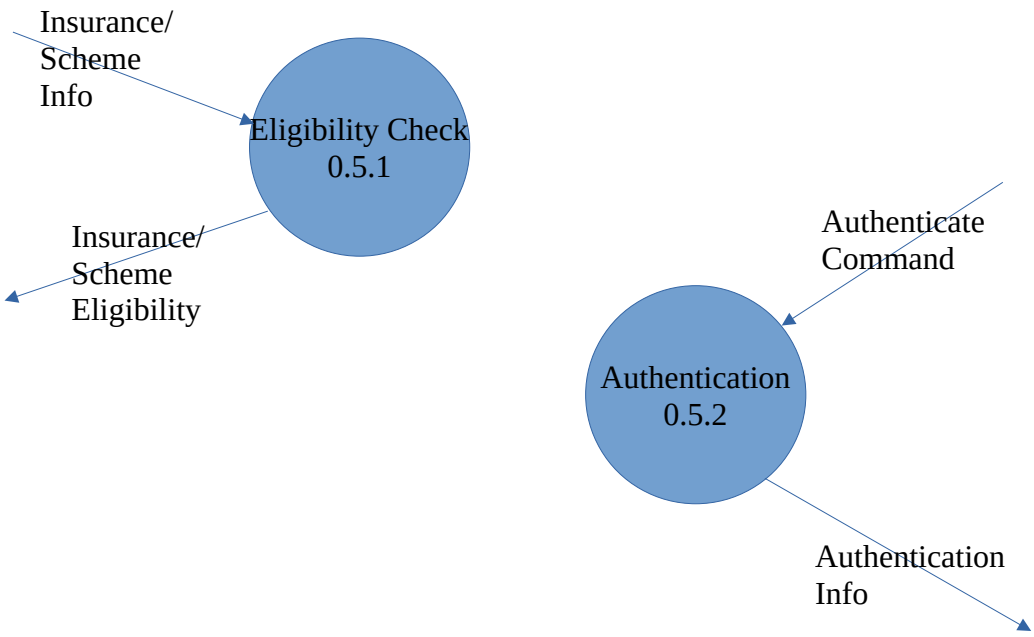
Payment Info
HIS Integration 0.4.3
Payment Record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Authentication
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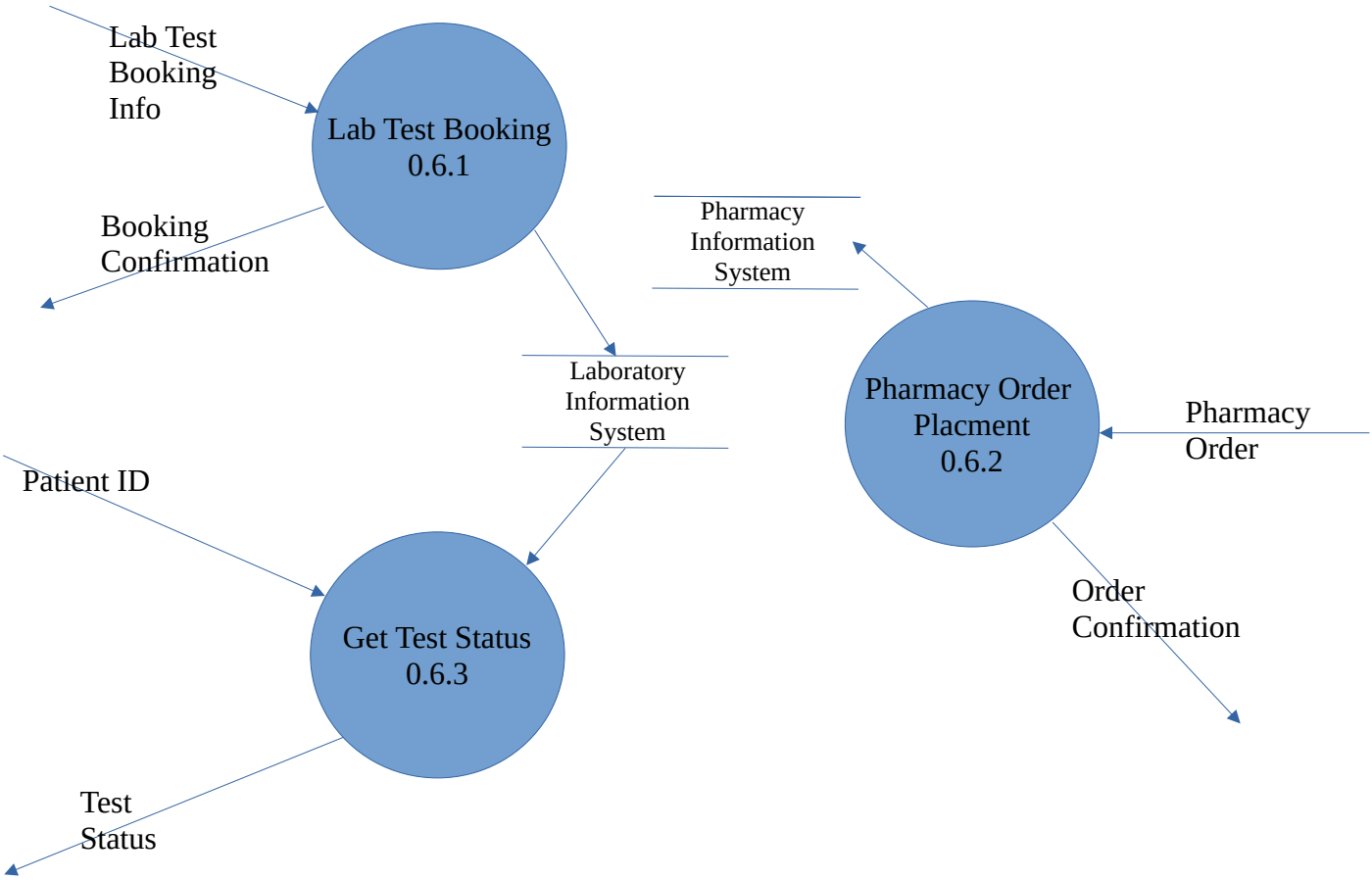
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Get Test Status
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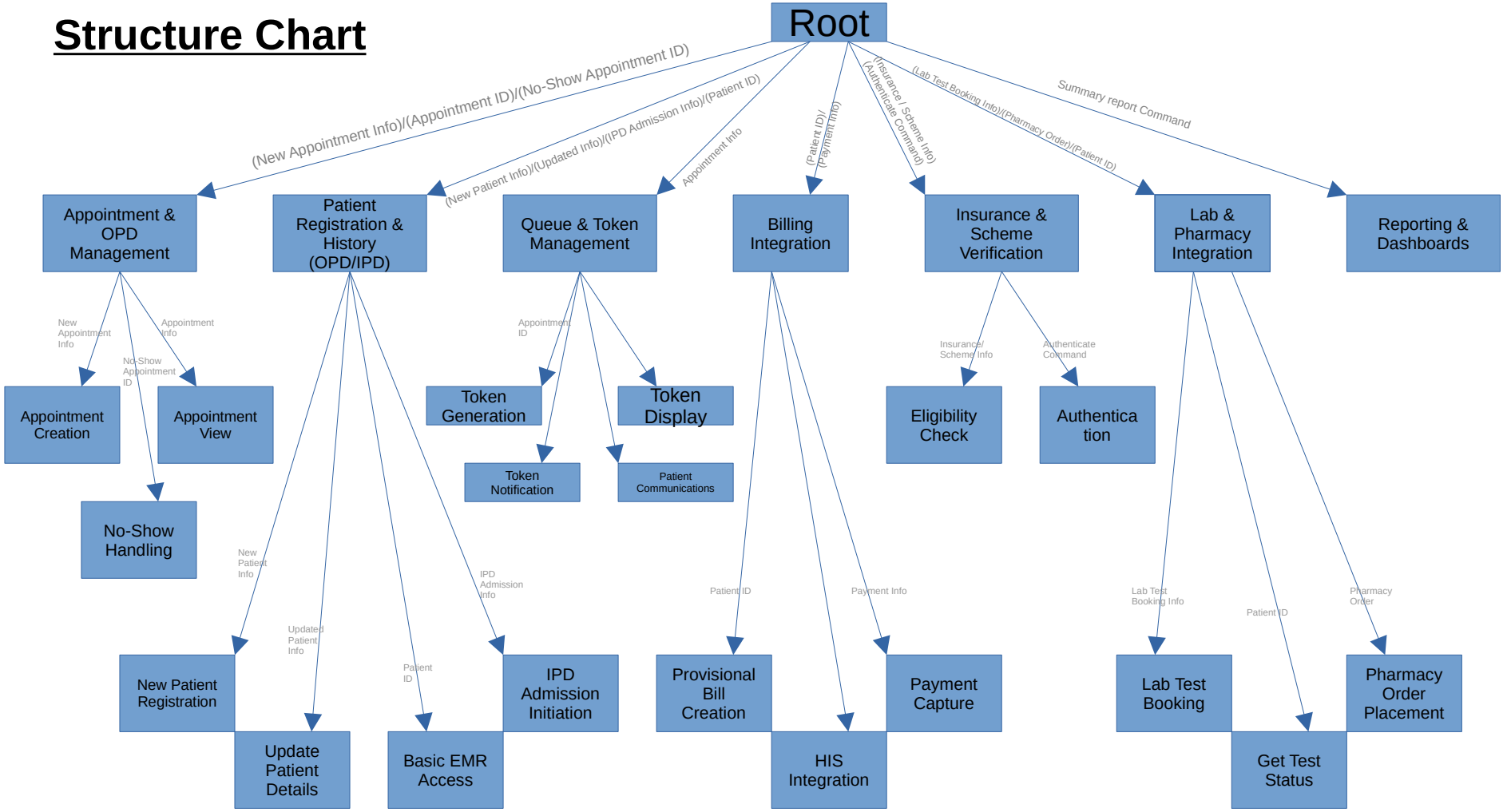
Order
Confirmation
Test
Status

Structure Chart



Data Dictionary

Data Decompositions :

```
Appointment_Info = {
    Appointment_ID,
    Patient_ID,
    Doctor_ID,
    Appointment_Date,
    Created_At
}
```

```
New_Appointment_Info = {
    Appointment_ID,
    Patient_ID,
    Doctor_ID,
    Appointment_Date
}
```

```
Token_Table = {
    Appointment_ID,
    Appointment_Time
}
```

```
Updated_Token_Table = {
    Appointment_ID,
    Appointment_Time
}
```

```
New_Patient_Info = {
    Patient_ID,
    Patient_First Name,
    Patient_Last_Name,
    Patient_Gender,
    Patient_Date_Of_Birth,
    Patient_Phone,
    Patient_Email,
    Patient_Address
}
```

```
Updated_Patient_Info = {
    Patient_ID,
    Patient_First Name,
    Patient_Last_Name,
    Patient_Gender,
    Patient_Date_Of_Birth,
    Patient_Phone,
    Patient_Email,
    Patient_Address
}
```

```
IPD_Admission_Info = {
    Admission_ID,
    Patient_ID,
    Admission_Date,
    Department,
    Doctor_Assigned,
    Admission_Reason,
    Initial_Diagnosis,
    Payment_Mode,
    Advance_Amount
}
```

```
Payment_Info = {
    Encounter_ID,
    Payment_Type,
    Transaction_Number,
    Amount,
}
```

```
Provisional_Bill = {
    Bill_ID,
    Encounter_ID,
    Amount,
    Created_At
}
```

```
Payment_Receipt = {
    Receipt_ID,
    Payment_ID,
    Bill_ID,
    Patient_ID,
    Payment_Date,
    Payment_Mode,
    Amount,
    Received_By,
    Remarks
}
```

```
Payment_Record = {
    Receipt_ID,
    Payment_ID,
    Bill_ID,
    Patient_ID,
    Payment_Date,
    Payment_Mode,
    Amount,
    Received_By,
    Remarks
}
```

```
Insurance_or_Scheme_Info = {
    Insurance_ID,
    Patient_ID,
    Coverage_Type,
    Provider_Name,
    Policy_Number,
    Scheme_Name,
    Valid_From,
    Valid_Till,
    Total_Coverage_Amount,
    Verification_Date,
}
```

```
Authentication_Info = {
    Auth_ID,
    Insurance_ID,
    Patient_ID,
    Request_Timestamp,
    Auth_Status,
    Verification_Timestamp
}
```

```
Patient_History = {
    History_ID,
    Patient_ID,
    Chronic_Diseases,
    Past_Surgeries,
    Allergies,
    Current_Medications,
    Last_Admission_ID,
    Last_Visit_Date
}
```

```
Lab_Test_Booking_Info = {
    Booking_ID,
    Patient_ID,
    Test_ID_List,
    Booking_Date,
    Scheduled_Date,
    Booking_Mode,
    Payment_Status
}
```

```
Pharmacy_Order = {
    Order_ID,
    Patient_ID,
    Ordered_Medicines,
    Order_Date,
    Order_Mode,
    Payment_Status,
    Dispensed_By,
    Dispense_Status,
    Total_Amount
}
```

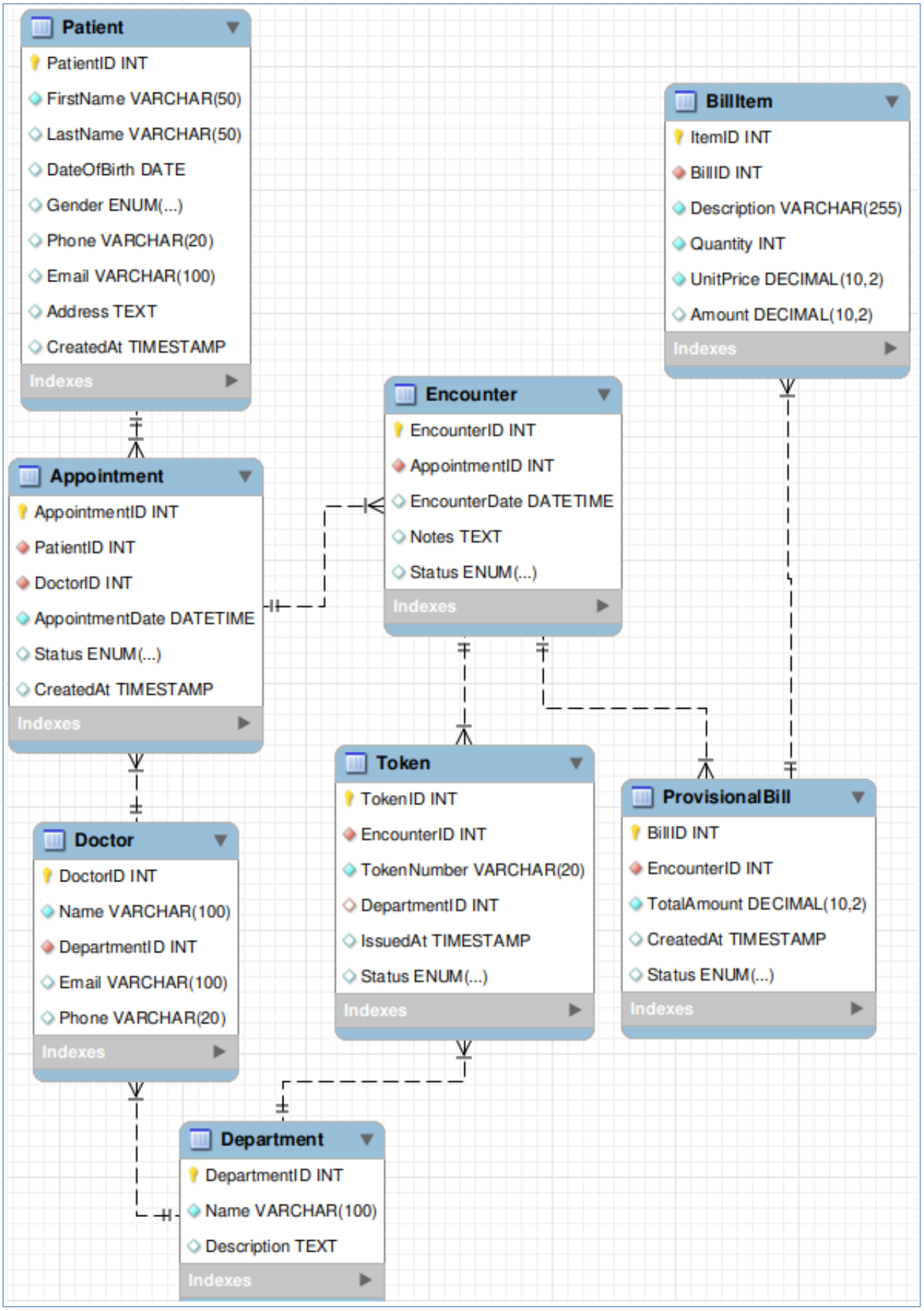
```
Test_Status = {
    Status_ID,
    Booking_ID,
    Test_ID,
    Sample_ID,
    Status,
    Status_Timestamp,
}
```

```
Summary_Report = {
    Report_ID,
    Report_Date,
    Department_ID,
    Doctor_ID,
    Total_Appointments,
    Completed_Encounters,
    Pending_Encounters,
    Cancelled_Appointments,
    Average_Wait_Time,
    Peak_Hours,
    First_Token_No,
    Last_Token_No,
}
```

Data Descriptions:

Data Name	Description	Type
Appointment_ID	Unique identifier for appointment	Integer
Patient_ID	Unique identifier for patient	Integer
Doctor_ID	Unique identifier for doctor	Integer
Appointment_Date	Date of appointment	DateTime
Created_At	Timestamp when appointment created	DateTime
No_Show_Appointment_ID	Identifier for missed appointment	Integer
Appointment_Time	Scheduled appointment time	Time
Patient_First_Name	Patient first name	String
Patient_Last_Name	Patient last name	String
Patient_Gender	Gender of patient	String
Patient_Date_Of_Birth	Patient date of birth	Date
Patient_Phone	Patient phone number	String
Patient_Email	Patient email address	String
Patient_Address	Patient residential address	String
Admission_ID	Unique identifier for hospital admission	Integer
Admission_Date	Date of hospital admission	Date
Department	Department handling the admission	String
Doctor_Assigned	Doctor assigned for the case	String
Admission_Reason	Reason for hospital admission	String
Initial_Diagnosis	Preliminary diagnosis at admission	String
Payment_Mode	Mode of payment (Cash/Card/Insurance)	String
Advance_Amount	Initial advance amount paid	Decimal
Encounter_ID	Unique identifier for patient encounter	Integer
Payment_Type	Type of payment (Cash/Card/Online)	String
Transaction_Number	Transaction reference number	String
Amount	Amount involved in transaction	Decimal
Bill_ID	Unique identifier for provisional bill	Integer
Created_At	Timestamp when bill created	DateTime
Receipt_ID	Unique identifier for receipt	Integer
Payment_ID	Identifier for payment transaction	Integer
Payment_Date	Date when payment made	Date
Received_By	Staff who received payment	String
Remarks	Additional comments or notes	String

Insurance_ID	Unique identifier for insurance/scheme record	Integer
Coverage_Type	Type of coverage (Private/Govt/Corporate)	String
Provider_Name	Name of insurance provider or scheme	String
Policy_Number	Insurance policy or card number	String
Scheme_Name	Name of insurance or health scheme	String
Valid_From	Start date of insurance validity	Date
Valid_Till	End date of insurance validity	Date
Total_Coverage_Amount	Maximum amount covered under policy	Decimal
Verification_Date	Date of policy verification	DateTime
Auth_ID	Unique identifier for authentication record	Integer
Request_Timestamp	Time when authentication requested	DateTime
Auth_Status	Result of authentication process	String
Verification_Timestamp	Time when authentication completed	DateTime
History_ID	Unique identifier for patient history record	Integer
Chronic_Diseases	List of chronic conditions	List<String>
Past_Surgeries	List of past surgical procedures	List<String>
Allergies	List of known allergies	List<String>
Current_Medications	List of current medications	List<String>
Last_Admission_ID	Reference to last hospital admission	Integer
Last_Visit_Date	Date of last hospital visit	Date
Booking_ID	Unique identifier for lab test booking	Integer
Test_ID_List	List of test identifiers booked	List<Integer>
Booking_Date	Date of booking	DateTime
Scheduled_Date	Scheduled date for sample collection	DateTime
Booking_Mode	Mode of booking (FrontDesk/Online/DoctorRequest)	String
Payment_Status	Status of payment for test	String
Order_ID	Unique identifier for pharmacy order	Integer
Ordered_Medicines	List of ordered medicines with quantity	List<Integer>
Order_Date	Date when order was placed	DateTime
Order_Mode	How order was placed	String
Dispensed_By	Staff who dispensed medicines	String
Dispense_Status	Current dispensing status	String
Total_Amount	Total cost of order	Decimal

Status_ID	Unique identifier for test status record	Integer
Test_ID	Reference to specific test	Integer
Sample_ID	Identifier for collected sample	Integer
Status	Current status of test	String
Status_Timestamp	Time when status last updated	DateTime
Report_ID	Unique identifier for report	Integer
Report_Date	Date of report generation	Date
Department_ID	Department for which report generated	Integer
Doctor_ID	Doctor for which report generated	Integer
Total_Appointments	Total appointments on report date	Integer
Completed_Encounters	Number of encounters completed	Integer
Pending_Encounters	Number of pending encounters	Integer
Cancelled_Appointments	Number of cancelled appointments	Integer
Average_Wait_Time	Average wait time for patients	Time
Peak_Hours	Time range with highest load	String
First_Token_No	First token number of the day	Integer
Last_Token_No	Last token number of the day	Integer

Entity Relationship Diagram